Lab - Research a Hardware Upgrade

Introduction

Use the Internet, a newspaper, or a local store to gather information about hardware components. Your customer's computer currently has one module of 2 GB of RAM, a 500 GB hard disk drive, and a PCIe video adapter card with 256 MB of RAM. Your customer wants to be able to play advanced video games.

Instructions

Step 1: Research memory options.

Shop around, and in the table below list the brand, model number, features, and cost for two different 8 GB modules of DDR3.

Brand and Model Number	Features	Cost
Kingston 8GB PC3-12800 CL11	*8GB, 1600MHZ, single memory module, voltage of 1.5V, 240 pins DDR3 SDRAM memory module, JEDEC standard 1.5V power supply, VDQQ = 1.5V*	*2,500*
Samsung M378B1G73EB0-YK0	*8GB, 240 pin, DIMM, UDIMM, JEDEC, 1.35V*	*3,963*

Question:

Based on your research, which RAM would you select? Be prepared to discuss your decisions regarding the RAM you select.

Kingston is much cheaper compared to Samsung, but Samsung is better when it comes to voltage because of its lower voltage which can run cooler and uses slightly less power. I will choose Kingston

Step 2: Research hard disk drive options.

Shop around, and in the table below list the brand, model number, features, and cost for a 3 TB 7200 rpm SATA 3 hard disk drive and a 500GB or 1TB solid state drive (SSD).

Brand and Model Number	Features	Cost
ADATA 1TB SU650	*1TB, 2.5", 3D NAND, sequential read up to 520MB/s, sequential write up to 450MB/s*	9,000
Seagate Barracuda ST3000DM001	3TB, 7200RPM, SATA 3, 3.5 inches, sequential up to 150MB/s to 180MB/s	4,450

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Question:

Based on your research, which hard disk drive would you select? Be prepared to discuss your decisions regarding the hard disk drive you select.

ADATA is significantly better because of its speed and reliability, Seagate has more storage capacity compared to ADATA but is very slow.

Step 1: Research video adapter card options.

Shop around, and in the table below list the brand, model number, features, and cost for two different PCIe video adapter cards with at least 1 GB of RAM.

Brand and Model Number	Features	Cost
MSI GeForce GT 1030	4GB Ram, 128-bit, boost clock up to 2321 MHz, GDDR4, 3840x2160, 4K UHD	6,165
ASRock AMD Radeon RX 6400 Challenger ITX	4GB Ram, 64-bit, boost clock up to 1665 MHz, GDDR4, 7680x4320,	7,398

Question:

Based on your research, which video adapter card would you select? Be prepared to discuss your decisions regarding the video adapter card you select.

AMD has better raw performance and efficiency in newer games compared to MSI

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